Analytical Cumulative Examination
4/20/07

Raman Spectroscopy/Light Scattering

1. (15) Draw an energy diagram for a molecular system showing the ground electronic state, the first excited state, and the transitions leading to absorption in the UV/Vis region, absorption in the IR region, fluorescence and Raman. Clearly label the diagram.

2. (10) Is the presence of a Rayleigh line evidence that the substance irradiated is Raman-active? Please explain your answer.

3. (20) A Raman-active transition in CCl_4 is observed at 459 cm^{-1}. If the sample is irradiated with a source at 488.0 nm, determine the wavelengths where the Stokes and anti-Stokes lines will be observed. Label each appropriately.

4. (15) Quantitatively determine the effect of changing the excitation wavelength of the source in a Raman spectrometer from 632.8 nm to 325.0 nm.

5. (15) A high resolution Raman spectrum of CCl_4 shows the transition at 459 cm^{-1} to be a triplet with a separation of 3 cm^{-1}. Determine the separation of this triplet in nanometers if excitation is at 488.0 nm.

6. (15) Briefly explain the basis for SERS (Surface Enhanced Raman Spectroscopy).

7. (10) Whereas most spectroscopic transitions are characterized by two basic quantities, frequency and intensity, Raman transitions are characterized by a third basic quantity, degree of polarization. Briefly discuss the basis for this characteristic and discuss what structural feature can be gleaned from measurement of this characteristic.

Cume April 20, 2007 - Spectroscopic Techniques

Spectroscopy is arguably the most important tool for investigating the structure of matter, and is also the basis of a number of important analytical methods. All spectroscopies involve the interaction of electromagnetic radiation with matter. Depending on the technique, a photon can be absorbed, a photon can be emitted, or a photon can be absorbed and another photon emitted. Because energy is conserved, the net photon energy stays in the molecule and changes its internal state. This can be a change in the electronic wavefunction, in which an electron is transferred from one orbital to another, a change in the vibrational wavefunction where the state of the nuclear vibrations change, a change in the rotational wavefunction (for freely rotating gas-phase molecules only), a change in the spin direction of the electrons, or a change in the spin direction of the nuclei. (Often several of these change simultaneously but the most important is the one that is associated with the largest energy change).

For each of the spectroscopic techniques described below, give

(a) A brief description of the technique (e.g. usually absorption, emission or absorption and emission)
(b) The main change in the internal status of the molecule
(c) A typical wavelength of the electromagnetic radiation used in this type of spectroscopy (in m or its standard multiples, e.g. μm).
(d) The frequency associated with the wavelength in (c), in units of Hz.
(e) The photon energy associated with the wavelength in (c), in units of J.

Constants: speed of light $c = 2.998\times10^{8}$ m/s, Planck's constant $h=6.626\times10^{-34}$ Js.
For your convenience, the first question is worked out.

1. Visible spectroscopy. (a) Usually measured in absorption, (b) It is associated with electronic transitions in molecules with low-lying electronic states, (c) 0.5 μm, (d) 6.0×10^{14} Hz (e) 3.97×10^{-19} J.

2. Infrared spectroscopy

3. Ultraviolet spectroscopy

4. NMR spectroscopy

5. Fluorescence spectroscopy

6. Microwave spectroscopy

7. X-ray photoelectron spectroscopy

8. Raman spectroscopy